

NAME SCHOOL TEACHER 

Pre-Junior Certificate Examination, 2020

Mathematics

Paper 1

Higher Level

Time: 2 hours, 30 minutes

300 marks

School Stamp

Running Total	
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For Examiner			
Question	Mark	Question	Mark
1		11	
2		12	
3		13	
4		14	
5			
6			
7			
8			
9			
10		Total	

Grade

Instructions

There are 14 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in the spaces provided in this booklet. You may lose marks if you do not do so. There is space for extra work at the back of the booklet. You may also ask the superintendent for more paper. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

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Question 1

(Suggested maximum time: 10 minutes)

- (a) A school held a fundraiser night to help raise money for two charities. They collected €2100 and divided it between the charities in the ratio 5:2.

How much did each charity receive?

[illegible]

1st charity received	€
----------------------	---

€	
---	--

2nd charity received | €

€	
---	--

- (b) Pat, a block layer, worked for 3.5 hours and laid 294 blocks.
- (i) How many blocks would he lay in 8 hours if he continued to work at the same speed?

- (ii) Pat is using a standard block which, including the mortar joint measures 450 mm by 225 mm when laid on its edge.
How many blocks laid on their edge will it take Pat to build a 50 m² wall?

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger square at the top left corner, likely for a title or header. The grid is empty and ready for use.

Question 2

(Suggested maximum time: 10 minutes)

Emma has a gross annual income of €45 000.

Emma pays income tax on her taxable income at a rate of 20% on the first €35 300, and 40% on the balance. She has annual tax credits of €1650.

- (a)** Work out Emma's net income, after income tax has been deducted.

[illegible]

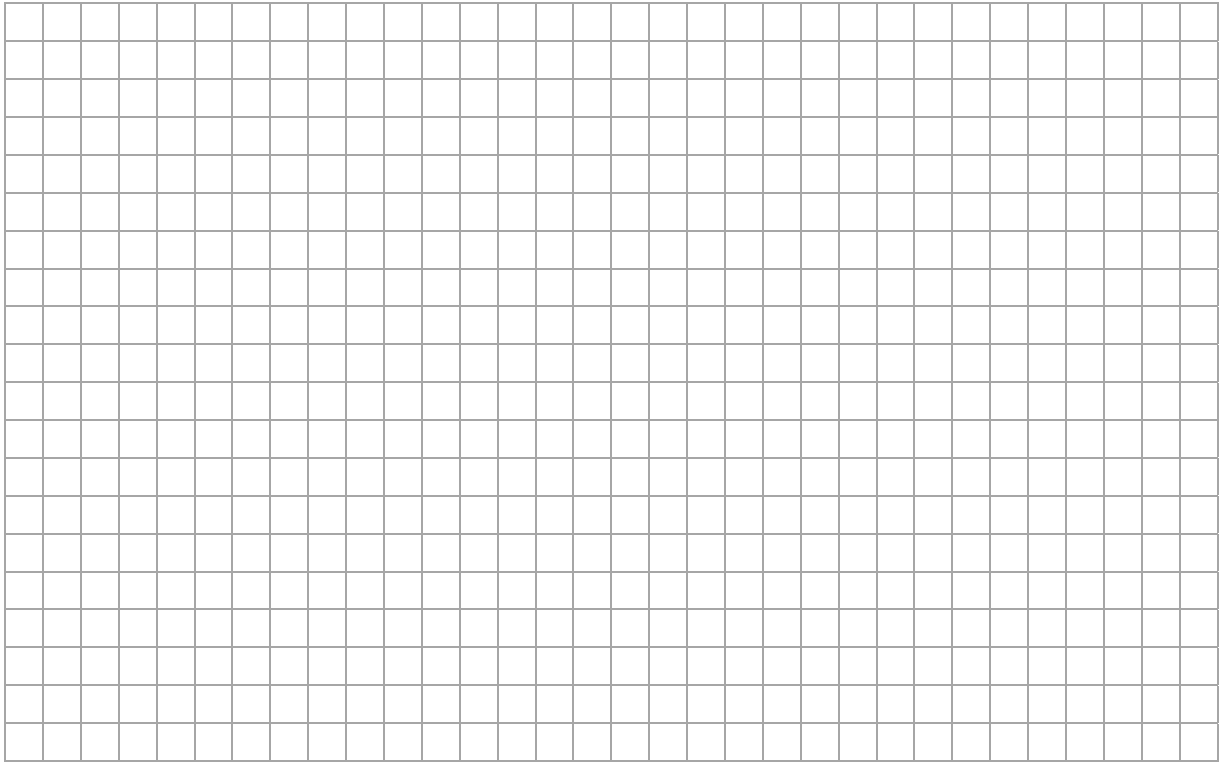
- (b)** Naya's net annual income, after income tax has been deducted, is €36 560. Naya pays income tax at the same rates and has the same annual tax credits as Emma. Work out Naya's **gross** annual income.

This image shows a full page of blank graph paper. The grid consists of small, uniform squares formed by thin, light gray lines. There are no margins, text, or other markings on the page.

Question 3**(Suggested maximum time: 10 minutes)**

- (a) Solve the following equation. Give each answer correct to two decimal places.

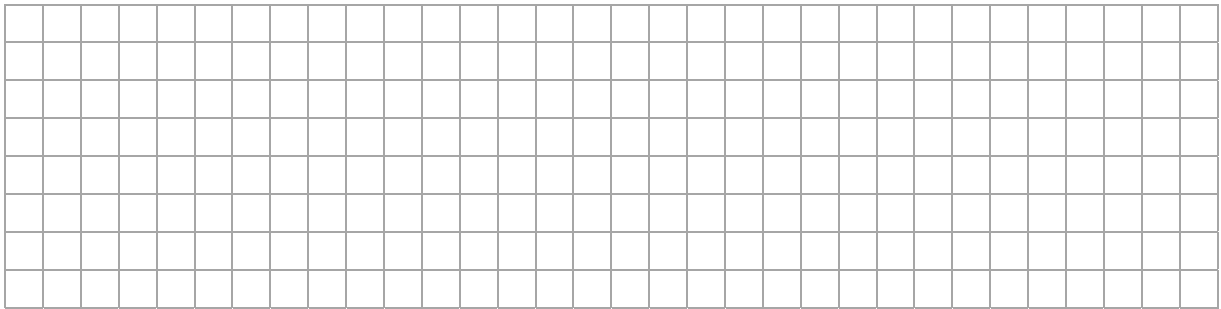
$$3x^2 + 5x - 13 = 0.$$



Answer:

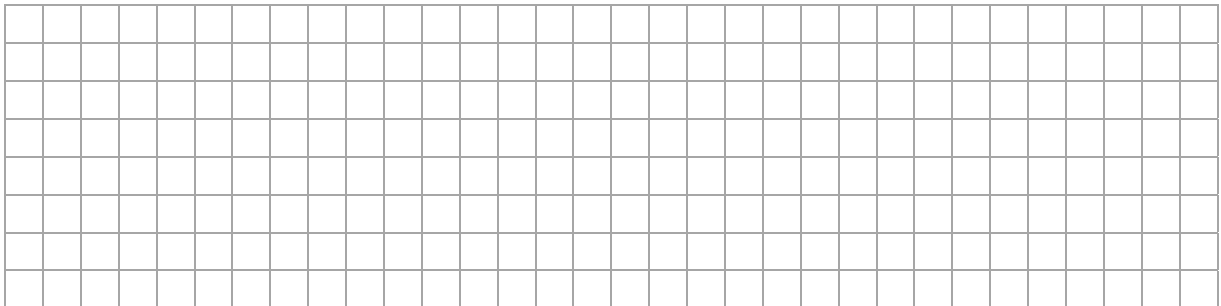
and

- (b) Multiply out and simplify $(x + 10)^2$.



- (c) Show that the following is always divisible by 5.

$$(x + 10)^2 - (x - 10)^2$$



Question 4

(Suggested maximum time: 10 minutes)

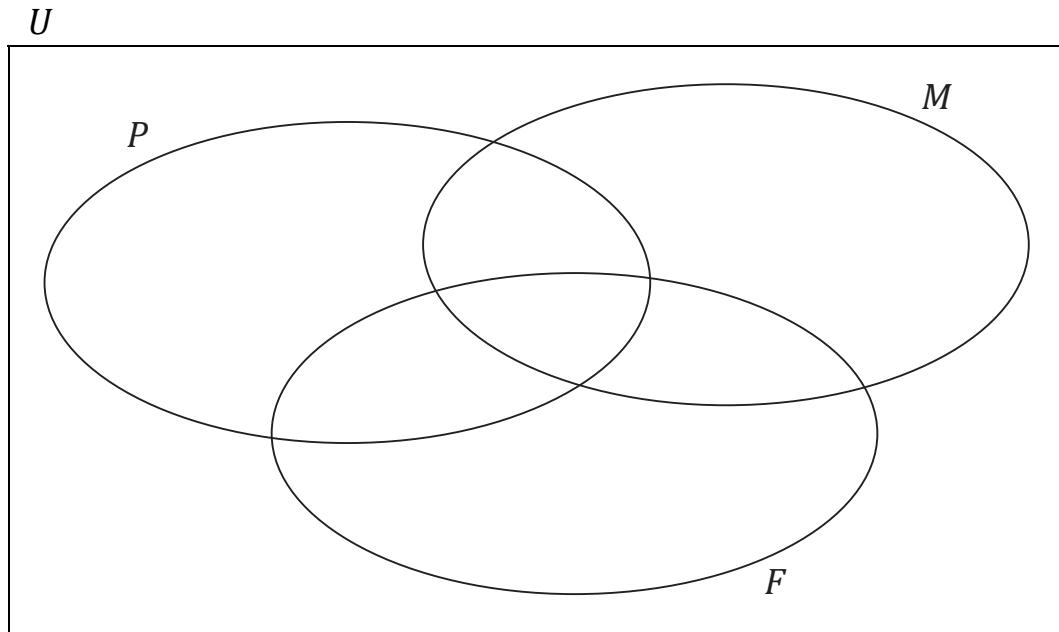
U is the set of natural numbers between 1 and 20 inclusive.

P is the set of prime numbers less than 20.

M is the set of multiples of 3 less than 20.

F is the set of factors of 20.

- (a)** Use this information to fill in the Venn diagram below.

[illegible]

- (b)** Use set notation to describe an empty set in the Venn diagram above.

[illegible]

- (c) Write down $\#(P \cap F)$.

[illegible]

- (d) List the elements of $(P \cup M \cup F)'$.

[illegible]

Question 5

(Suggested maximum time: 10 minutes)

- (a)** What is a natural number?

[illegible]

- (b)** The table below shows whether each of the given numbers is an element of the natural numbers ($\mathbb{N}$), the integer numbers ($\mathbb{Z}$), the rational numbers ($\mathbb{Q}$) and the irrational numbers ($\mathbb{R} \setminus \mathbb{Q}$).

Complete the table by writing “Yes” or “No” into each box.

The first row is already done for you: 19 is an element of $\mathbb{N}$, $\mathbb{Z}$, and $\mathbb{Q}$ but not $\mathbb{R} \setminus \mathbb{Q}$.

Number	Number Sets			
	$\mathbb{N}$	$\mathbb{Z}$	$\mathbb{Q}$	$\mathbb{R} \setminus \mathbb{Q}$
19	Yes	Yes	Yes	No
π				
0.123				
$\sqrt{3}$				
-7				
$-\frac{1}{3}$				

[illegible]

- (c)** Use the properties of surds to simplify the following: $\sqrt{75} + \sqrt{48} - \sqrt{27}$. Show your workings and give your answer in the form $n\sqrt{3}$, where $n \in \mathbb{Z}$.

A large grid of graph paper with 20 columns and 10 rows. The grid is composed of small squares, with a slightly larger margin on the left side for writing.

Question 6

(Suggested maximum time: 10 minutes)

Susan and Jill decided to join their local 5k park run on a Saturday morning.

- (a) Susan ran her first 5 km run in 30 minutes. For a number of consecutive weeks, she decreased her time each week by 2 minutes.

- (i) Work out Susan's **time** taken to complete the 5 km run on the **6th week**.

[illegible]

- (ii) Write an expression in n for the **time** taken by Susan to complete the 5 km run on the **n th week**.

Susan's time on the n th week:

minutes

[illegible]

- (b) Jill ran her first 5 km run in 27 minutes. For a number of consecutive weeks, she decreased her time each week by 1 minute. Write an expression in n for the **time** taken by Jill to complete the 5 km run on the **n th week**.

Jill's time on the n th week:

--

minutes

[illegible]

- (c)** Using your expressions from part **(a)** and **(b)**, or otherwise, work out which week Susan and Jill finished the park run at the **same time**.

[illegible]

Question 7

(Suggested maximum time: 10 minutes)

- (a)** Factorise fully $8x^2 - 50y^2$.

[illegible]

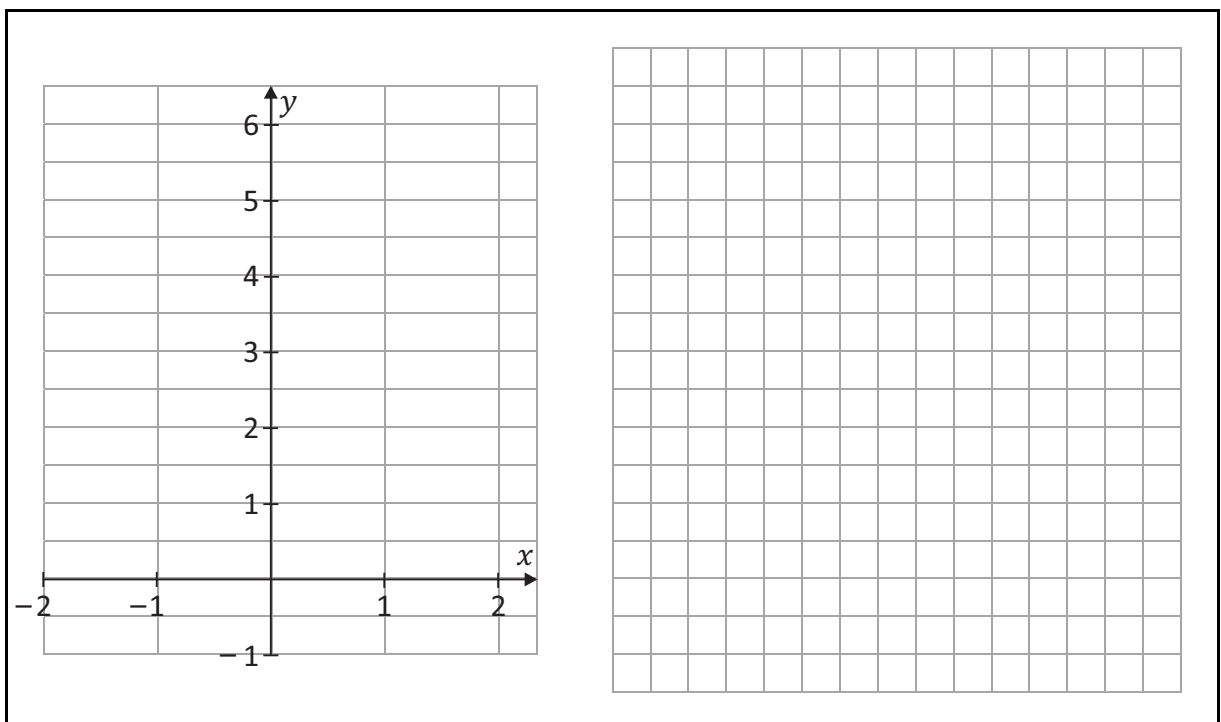
- (b)** Factorise fully $6kn + km - 2k^2 - 3mn$.

[illegible]

- (c) (i)** Factorise fully $2x^2 - 3x + 1$

[illegible]

- (ii) Draw the function $y = 2x^2 - 3x + 1$ in the domain $-1 \leq x \leq 2$, for $x \in \mathbb{R}$. Show your working out.



Question 8 (Suggested maximum time: 10 minutes)

Question 8 (Suggested maximum time: 10 minutes)

Water used in m ³	Cost in €
100	300
200	450
300	600
400	750
500	900
600	1050

- (a) Put a tick in the correct box to show what type of relationship exists between the volume of water used and the cost. Tick (✓) **one** box only. **Justify** your answer.

linear

quadratic

1

exponential

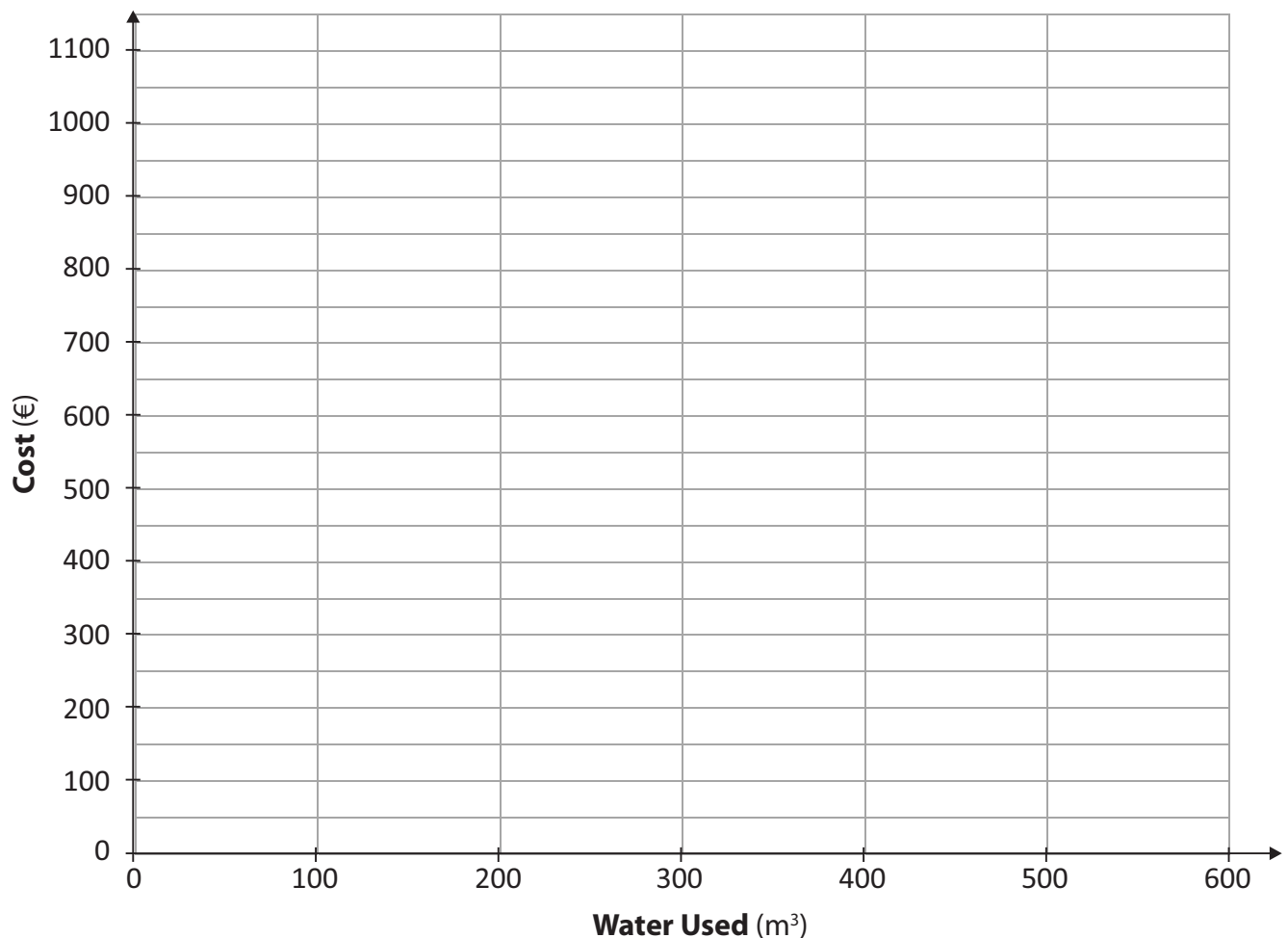
5

none of these

5

Justification:

- (b)** Using the figures in the table above, **draw** a graph to show the relationship between the volume of water used and the cost.



(c) What is the cost of the standing charge?

[illegible]

(d) Write down a formula to represent the relationship between the number of units of water used and the cost for any given number of units.

[illegible]

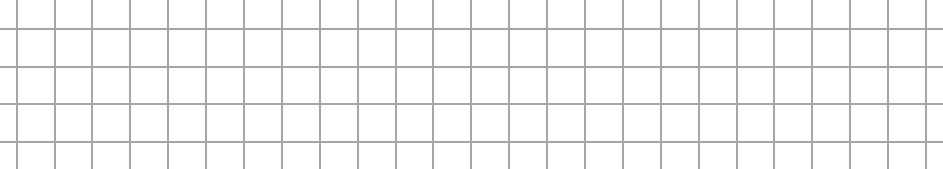
Question 9 (Suggested maximum time: 10 minutes)

The furthest possible distance between Earth and Mars, based on their orbit around the sun is 401 000 000 km.

- (a)** Write this distance in the form $a \times 10^n$ km, where $1 \leq a \leq 10$ and $n \in \mathbb{Z}$.

[illegible]

- (b)** The closest Earth and Mars get to each other is 5.45×10^7 km. What is the **percentage increase** between this distance and the distance from part **(a)** above? Give your answer correct to two decimal places.



- (c) The most recent Spacecraft sent from Earth to land successfully on Mars took 205 days. It travelled a total of 2×10^8 km. Find the **average speed** of the spacecraft in km/hr. Give your answer, correct to two significant numbers, in the form $a \times 10^n$ km, where $1 \leq a \leq 10$ and $n \in \mathbb{Z}$.

[illegible]

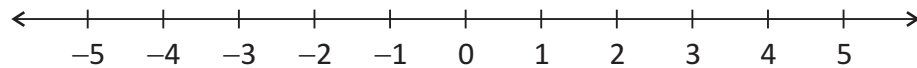
Question 10

(Suggested maximum time: 10 minutes)

- (a)** Use algebra to solve the following inequality:

$$-14 \leq 3x - 5 < 1, x \in \mathbb{Z}.$$

- (b)** Graph your solution to part **(a)** on the number line given.



- (c)** In Ireland, road distance is measured in kilometres. In the UK, road distance is measured in miles. Emma wants to convert a distance from kilometres (k) to miles (m). She multiplies the distance by 5 and then divides by 8.

- (i) Using the information above, convert 320 kilometres to miles.

[illegible]

- (ii) Write an algebraic expression that converts kilometres (k) to miles (m).

$$m =$$

--	--

- (iii) Write another algebraic expression that converts miles (m) to kilometres (k).

$$k =$$

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- (iv)** Use your expression in part **(iii)** to verify your answer in part **(i)**.

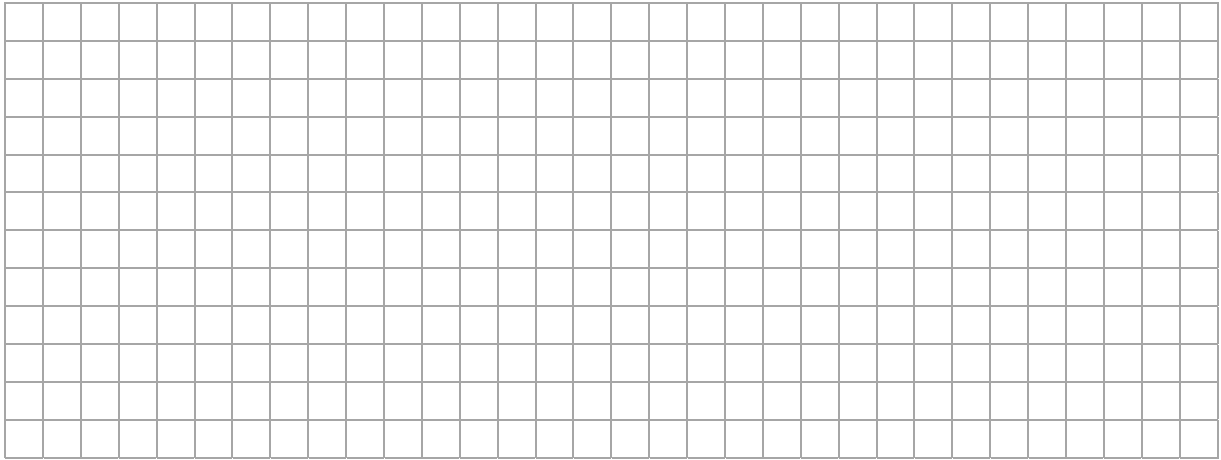
[illegible]

Question 11

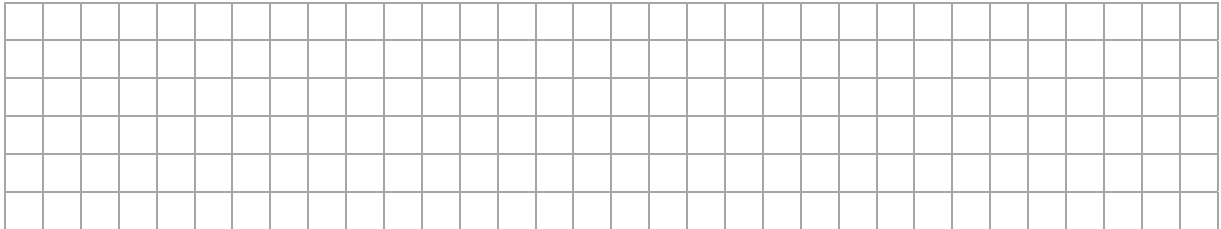
(Suggested maximum time: 10 minutes)

- (a)** Solve the simultaneous equations:

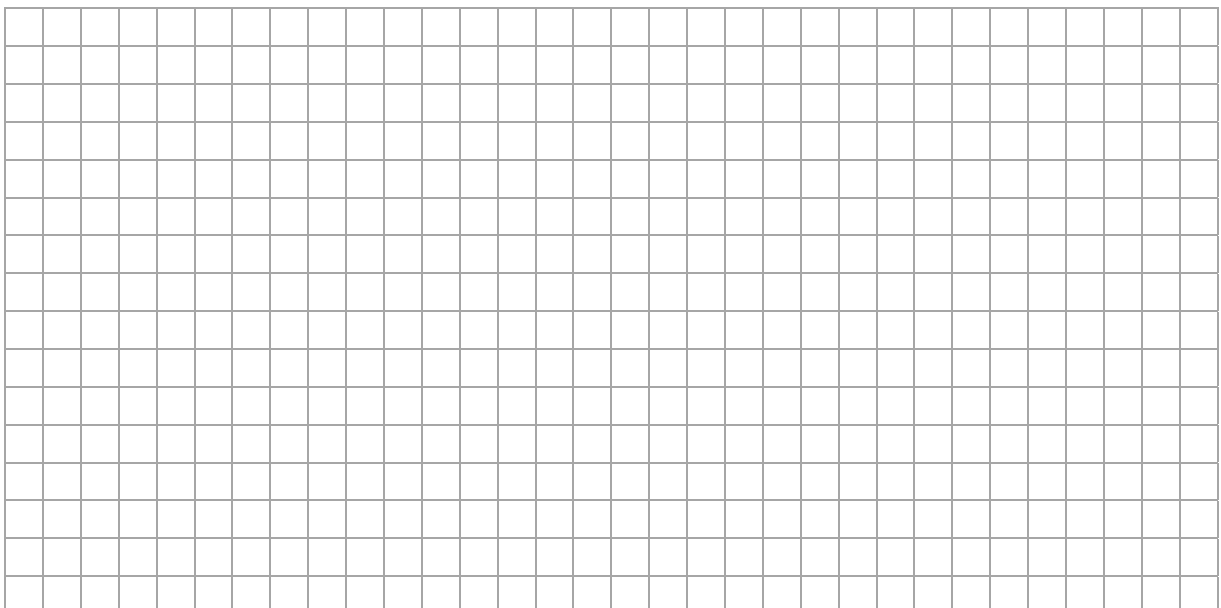
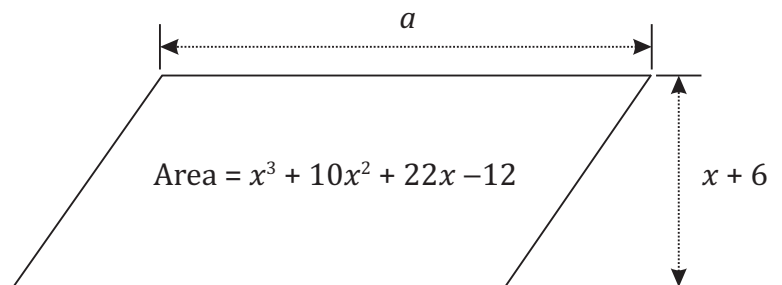
$$\begin{aligned} 3x + 4y &= 7 \\ 5x - 2y &= 16. \end{aligned}$$



- (b)** Verify your answer to **(a)**.



- (c)** The **area** of the parallelogram shown is $x^3 + 10x^2 + 22x - 12$ square units. It has a side of length $x + 6$ units. Find the length of the side marked a in the diagram.



Question 12 (Suggested maximum time: 10 minutes)

Question 12 (Suggested maximum time: 10 minutes)

The following number pattern forms a quadratic sequence.

3, 8, 15, ...

- (a)** Work out the next three terms of the sequence.

[illegible]

3,	8,	15,		,		,
----	----	-----	--	---	--	---

- (b)** Find a formula for the n th term of this sequence, where $n \in \mathbb{N}$.

n th term:

--

- (c)** Show that 195 is a term in this sequence, if the pattern continues.

Question 13 (Suggested maximum time: 5 minutes)

Question 13 (Suggested maximum time: 5 minutes)

A Fitbit costs €180 which includes VAT at a rate of 23%.

- (a)** Work out the cost of the Fitbit before VAT?
Give your answer correct to the nearest cent.

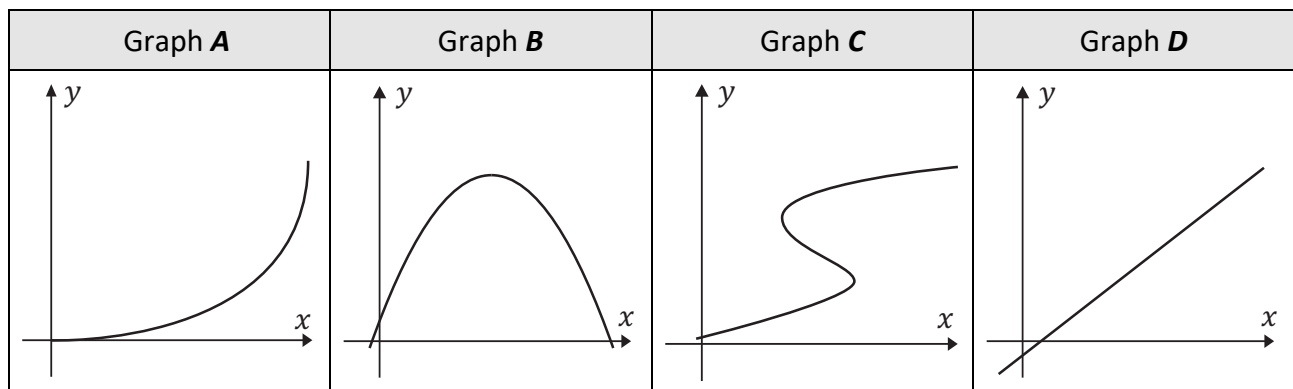
- (b)** In the UK, the same Fitbit costs £160. If the exchange rate is €1 = £0.86. Is the Fitbit cheaper to purchase in Ireland or in the UK? Justify your answer.

[illegible]

Question 14

(Suggested maximum time: 10 minutes)

- (a)** The table below shows four graphs, **A**, **B**, **C**, and **D**.



Put a tick in the correct box to show which of these graphs is **not** a function.

Tick (✓) **one** box only. **Justify** your answer.

A

B

C

D

1

Justification:

- (b)** A function is defined as $f(x) = 4x - 3$, where $x \in \mathbb{R}$.

- (i) Find the value of $f(5)$.

[illegible]

- (ii) Write $f(k)$ and $f(2k - 3)$ in terms of k .

[illegible]

- (iii) Hence, find the value of k for which $f(k) = f(2k - 3)$.

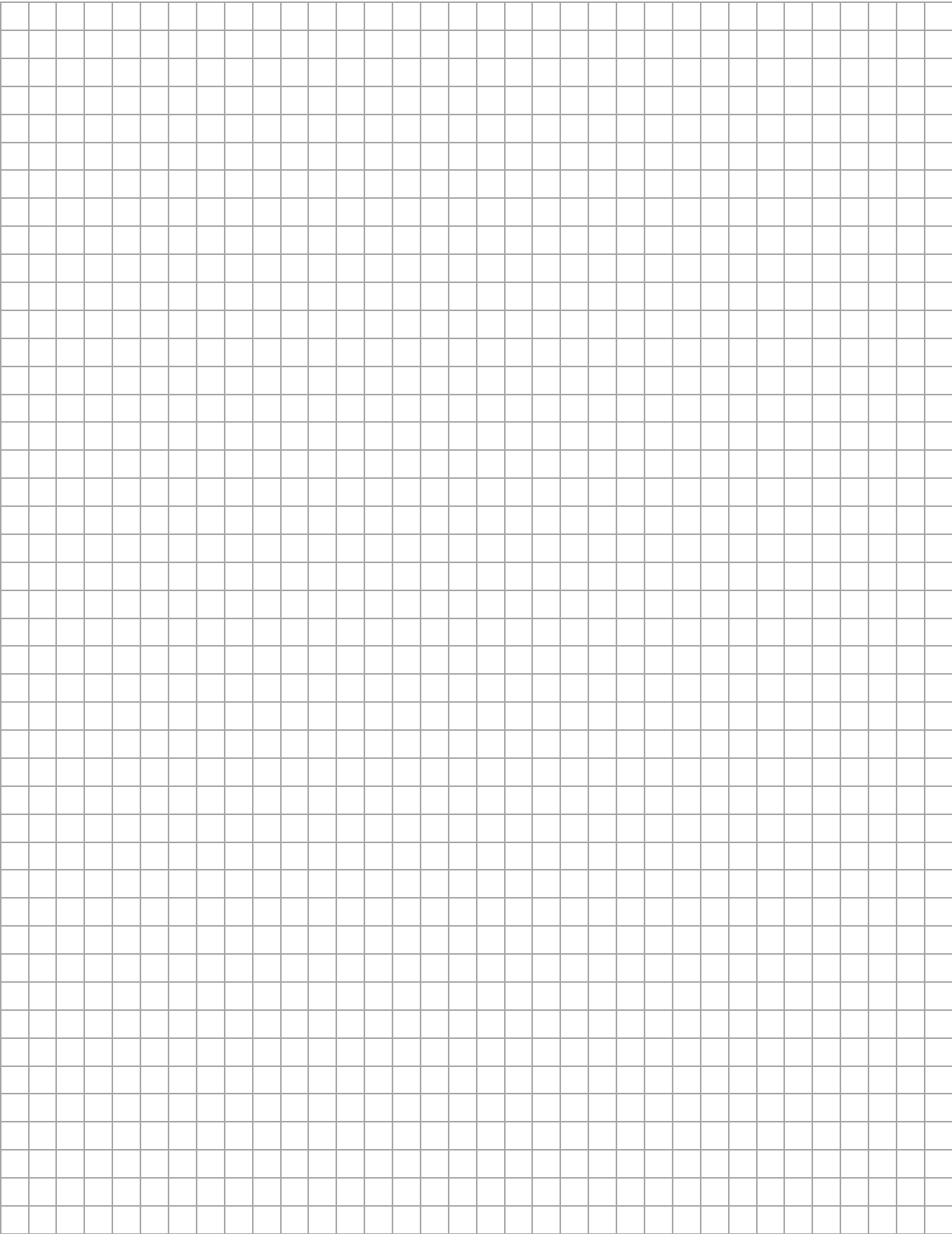
[illegible]

Page for extra work.

Label any extra work clearly with the question number and part.

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Pre-Junior Certificate Examination, 2020

Mathematics – Paper 1

Higher Level

Time: 2 hours, 30 minutes

